Hoang Trieu Vy LE

lehoangtrieuvy15@gmail.com
Hoang Trieu Vy LE
hoangtrieuvy.github.io

SUMMARY — I am a Senior Research Scientist at Beink Dream, where I design and develop *Beink ADN*, a 3D/CAD copilot, and act as solution architect for the company's AI systems. My work focuses on multi-modal understanding across 2D/3D/CAD scenes — with applications in computer-aided design, robotics and human modelling. Prior to that, I was a Research Engineer at CEA, and I completed my Ph.D. at École Normale Supérieure de Lyon under the supervision of Prof. Nelly Pustelnik, where I worked on unrolled proximal architectures bridging signal processing and deep learning for image restoration and interface detection.

EXPERIENCES

Beink Dream — **Mar 2025 – Present**
Senior Research Scientist — *Paris, France*

- **3D/CAD copilot** Conceived and built an agentic copilot coupling MLLM reasoning with generative vision models and geometry-aware constraints that turns sketches, images and natural-language intent into structured 3D and parametric CAD output.
- **Solution architect for AI systems.** Defined the end-to-end architecture of the company's AI stack — model selection, agentic orchestration, inference and serving, evaluation harnesses — and drove the path from research prototype to production at scale.

CEA – DIGITEO Labs — **Feb 2024 – Feb 2025**
Research Engineer — *Saclay, France*

Lightweight learning models for 3D CT reconstruction — physics-consistent volumetric scene representations recovered from sparse, noisy and limited high-dimensional measurements.

INMA, ICTEAM – UC Louvain — **Sep 2020 – Aug 2021**
Visiting PhD Researcher — *Louvain, Belgium*

Designed scalable accelerated algorithms to solve joint image restoration and edge detection on large-scale physics experiment images.

OGoxe – Météo France — **Jan 2020 – Aug 2020**
Research Intern — *Toulouse, France*

Built velocity estimation models from video sequences — inferring physical dynamics (fluid flow) from continuous visual observations, an early instance of physics-grounded world dynamics modelling.

IMFT – Toulouse Fluid Mechanics Institute — **Jul 2019 – Oct 2019**
Research Intern – High Performance Computing — *Toulouse, France*

Novel bucketing-based algorithm for particle-based simulation using mesh-less models across GPU clusters.

EDUCATION

École Normale Supérieure de Lyon — **Sep 2020 – Dec 2023**
Ph.D. in Applied Mathematics, AI, Signal and Image Processing — *Lyon, France*

Thesis: *Variations on the Mumford-Shah functional for interface detection in degraded images: from proximal algorithms to unrolled architectures.*
Advised by Prof. Nelly Pustelnik & Dr. Marion Foare.

INSA Toulouse — **Sep 2015 – Sep 2020**
Master of Engineering in Applied Mathematics — *Toulouse, France*

Major in Mathematical Modelling and Numerical Analysis.

PUBLICATIONS

Unfolded Discrete Mumford-Shah Functional for Joint Image Denoising and Edge Detection, *EUSIPCO 2025*
H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Embedding Blake-Zisserman Regularization in Unfolded Proximal Neural Networks for Enhanced Edge Detection, *IEEE SPL 2025*
H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Proximal Neural Networks based Reconstruction for Few-View CT Applications, *iCT 2025*
H.T.V Le, C. Bossuyt, J. Escoda, H. Wang, J. De Beenhouwer, J. Sijbers

Unfolded Proximal Neural Networks for Robust Image Gaussian Denoising, *IEEE TIP 2024*
H.T.V Le, Audrey Repetti, Nelly Pustelnik

Deep Image Prior for Sparse-View Reconstruction in Static, Rectangular Multi-Source X-ray CT Systems for Cargo Scanning, *SPIE Developments in X-Ray Tomography XV, 2024*
C. Bossuyt, J. De Beenhouwer, J. Sijbers, D. Iuso, M. Costin, J. Escoda, H.T.V Le, A. J. den Dekker

Proximal Based Strategies for Solving Discrete Mumford-Shah With Ambrosio-Tortorelli Penalization on Edges, *IEEE SPL 2022*
H.T.V Le, Marion Foare, Nelly Pustelnik

The Faster Proximal Algorithm, the Better Unfolded Deep Learning Architecture? A Study Case of Image Denoising, *EUSIPCO 2022*
H.T.V Le, Nelly Pustelnik, Marion Foare

Fast Unfolded Proximal Algorithms for the Analysis of Piecewise Homogeneous Fractal Images, *Gretsi 2022*
H.T.V Le, Barbara Pascal, Nelly Pustelnik, Marion Foare, Patrice Abry

ACTIVITIES

Teaching: Signal processing, convex optimization, unfolded NNs and inverse problems (ENS Lyon); Signal processing and Cryptography Mathematics (EPITA Lyon); Fundamental Machine Learning (EPITA Paris)

Talks:
EUSIPCO, Belgrade, Serbia **Aug 2022**
Conference on Digital Geometry and Discrete Variational Calculus **Mar 2021**

Projects & Open-Source Software:

Zingu [Python, Embedded] — Two-wheeled self-balancing robot that recovers autonomously after a fall; ongoing work on the perception and decision stack.
Gist [PyTorch] — Compact scene representation from a single moving camera for navigation, recognition and change detection, without rendering.
Meepo [Python] — Local, private, always-on personal AI assistant with persistent long-term user memory.
Proximal Neural Networks [PyTorch] — Lightweight unrolled architectures combining iterative algorithms and deep networks for inverse problems.
P-DMS-AT [Python] — Proximal strategies for Discrete Mumford-Shah with Ambrosio-Tortorelli edge penalization.
BZ-PNN [PyTorch] — Super-lightweight model for enhanced edge detection via Blake-Zisserman regularization.
DMS-PNN [PyTorch] — Unfolded Discrete-Mumford-Shah Proximal Neural Networks.
Bucketing-based Algorithm [Fortran, C/C++, Python] — Parallelized linearly transformed particle method for diffusion equations.

AWARDS

- Third prize, Vietnam National Physics Olympiad **2015**
- Fourth prize, Vietnam National Physics Olympiad **2014**
- Silver medal, Northern Regional Vietnam Physics Olympiad **2013**